

# Fundamentals of Linear Control:

## A Concise Approach

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**Chapter 6: Controller Design**

[linearcontrol.info/fundamentals](http://linearcontrol.info/fundamentals)

# 6.1 Second-Order Systems

## Characteristic equation

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0, \quad \omega_n > 0,$$

- $\zeta$ : damping ratio
- $\omega_n$ : natural frequency

## Roots

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = (s - s_1)(s - s_2), \quad s_1 = -\omega_n(\zeta + \sqrt{\zeta^2 - 1}), \quad s_2 = -\omega_n(\zeta - \sqrt{\zeta^2 - 1})$$

## Real roots

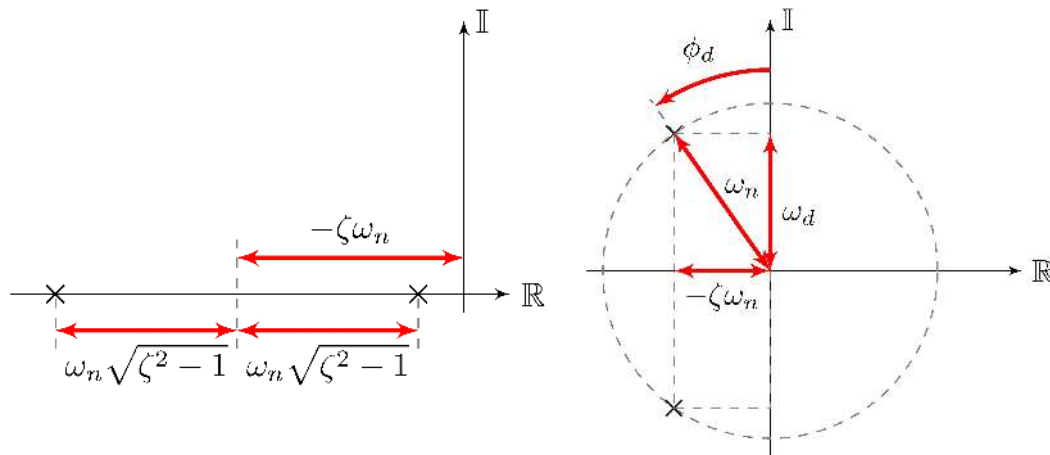
$$|\zeta| \geq 1$$

## Complex roots

$$|\zeta| < 1$$

# Second-Order Systems

real roots,  $|\zeta| \geq 1$     complex roots,  $|\zeta| < 1$



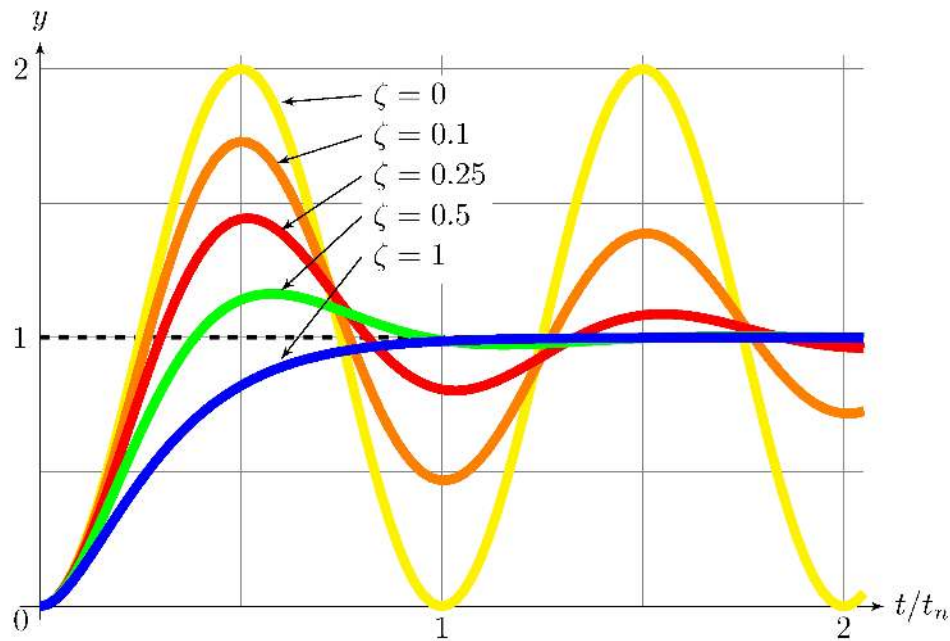
## Roots

$$s = -\omega_n(\zeta \pm \sqrt{\zeta^2 - 1}), \quad \omega_d = \omega_n \sqrt{1 - \zeta^2}, \quad \phi_d = \sin^{-1} \frac{\zeta \omega_n}{\omega_n} = \tan^{-1} \frac{\zeta \omega_n}{\omega_d} = \sin^{-1} \zeta$$

## Asymptotic stability

$$\zeta > 0$$

# Second-Order Systems

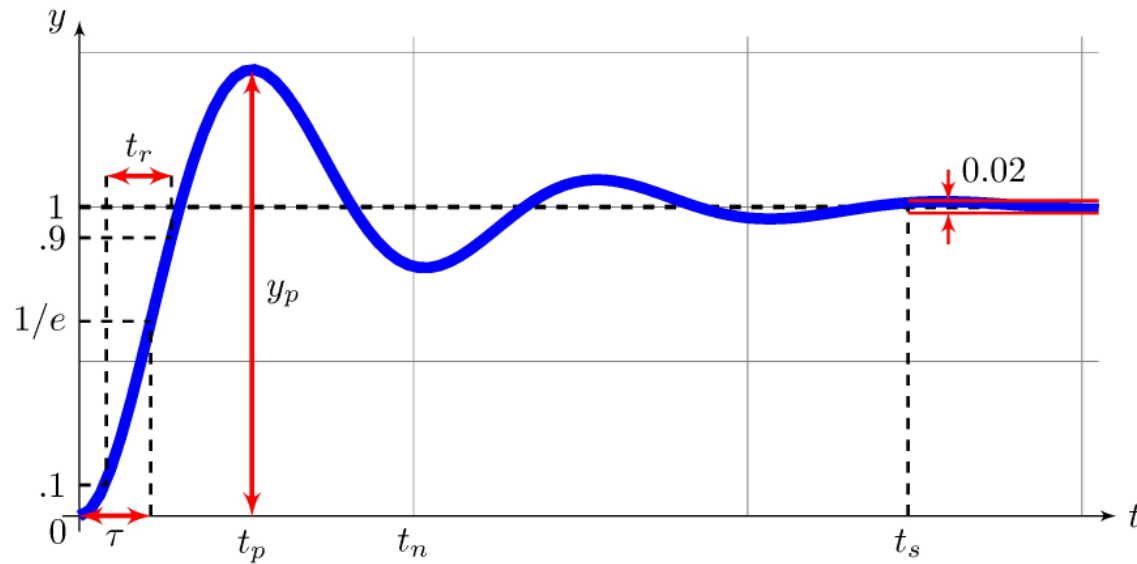


## Step response

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}, \quad y(t) = 1 - \frac{1}{\sqrt{1 - \zeta^2}} e^{-\zeta\omega_n t} \sin(\omega_d t + \pi/2 - \phi_d), \quad t \geq 0$$

# Second-Order Systems

## Step response

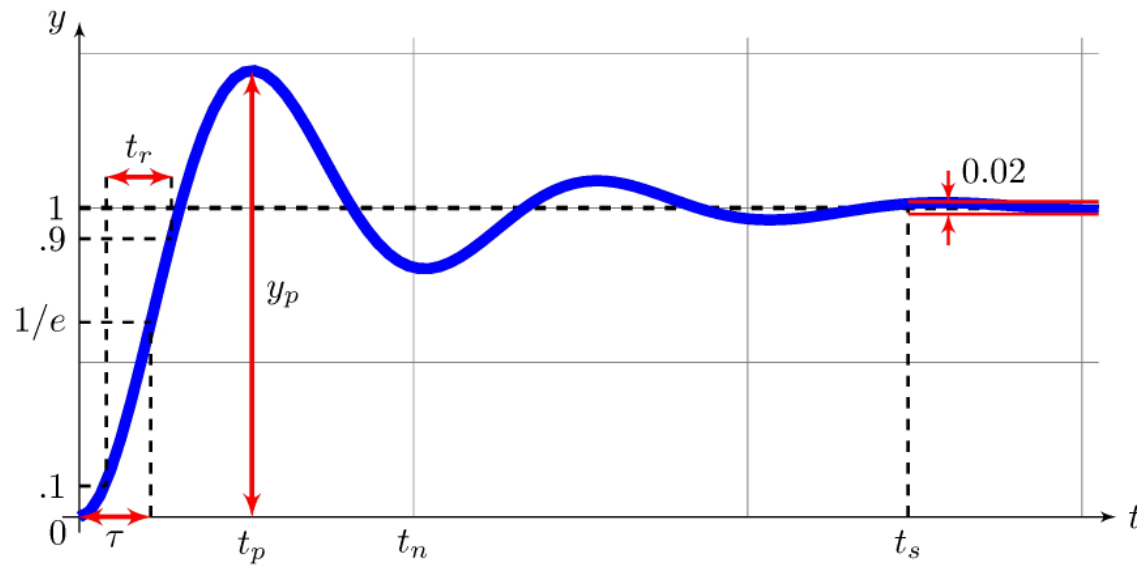


## Time-constant and Rise-time

$$\frac{t_r}{t_n} \approx 0.16 + 0.14\zeta + 0.24\zeta^3, \quad \frac{\tau}{t_n} \approx 0.19 + 0.1\zeta + 0.054\zeta^3, \quad t_n = \frac{2\pi}{\omega_n}$$

# Second-Order Systems

## Step response



## Overshoot and Settling-time

$$t_p = \frac{\pi}{\omega_d}, \quad y_p = 1 + e^{-\zeta\pi/\sqrt{1-\zeta^2}}, \quad t_s = \frac{\log(50)}{\zeta\omega_n} \approx \frac{3.9}{\zeta\omega_n}$$

# Example: car model with integral feedback

## Open-loop

$$G(s) = \frac{\frac{p}{m}}{s + \frac{b}{m}}, \quad K(s) = \frac{K_i}{s}$$

## Closed-loop

$$H(s) = \frac{K(s)G(s)}{1 + K(s)G(s)} = \frac{\frac{p}{m} K_i}{s^2 + \frac{b}{m} s + \frac{p}{m} K_i}$$

## Second-order

$$\omega_n = \sqrt{\frac{p}{m} K_i}, \quad \zeta = \frac{b/m}{2\sqrt{(p/m) K_i}} = \frac{b}{2\sqrt{p m K_i}}$$

# Example: car model with proportional-integral feedback

## Open-loop

$$G(s) = \frac{\frac{p}{m}}{s + \frac{b}{m}}, \quad K(s) = K_p + \frac{K_i}{s}$$

## Closed-loop

$$H(s) = \frac{K(s)G(s)}{1 + K(s)G(s)} = \frac{\frac{p}{m}(K_p s + K_i)}{s^2 + (\frac{b}{m} + \frac{p}{m}K_p)s + \frac{p}{m}K_i}$$

## Second-order

$$\omega_n = \sqrt{\frac{p}{m}K_i}, \quad \zeta = \frac{b/m + p/mK_p}{2\sqrt{(p/m)K_i}} = \frac{b + pK_p}{2\sqrt{p m K_i}}$$



## Example: simple pendulum

### Open-loop (stable equilibrium)

$$G_0(s) = \frac{1/J_r}{s^2 + (b/J_r)s + m g r/J_r}$$

### Second-order

$$\omega_{n_{ol}} = \sqrt{\frac{m g r}{J_r}}, \quad \zeta_{ol} = \frac{b}{2\sqrt{J_r m g r}}$$

### Open-loop (unstable equilibrium)

$$G_\pi(s) = \frac{1/J_r}{s^2 + (b/J_r)s - m g r/J_r}$$

# Example: simple pendulum

## Open-loop

$$G_{\pi}(s) = \frac{1}{J_r s^2 + b s - m g r}$$

## Proportional feedback (unstable equilibrium)

$$H_{\pi}(s) = \frac{K G_{\pi}(s)}{1 + K G_{\pi}(s)} = \frac{K}{J_r s^2 + b s + K - m g r},$$

$$\omega_{n_{\pi}} = \sqrt{\frac{K - m g r}{J_r}}, \quad \zeta_{\pi} = \frac{b}{2\sqrt{J_r(K - m g r)}}$$

## Closed-loop stability

$K > m g r$ , for example

$$K = 3m g r \quad \implies \quad \omega_{n_{\pi}} = \sqrt{2} \omega_{n_{ol}}, \quad \zeta_{\pi} = \frac{1}{\sqrt{2}} \zeta_{ol}$$

# Example: simple pendulum

## Open-loop

$$G_0(s) = \frac{1}{J_r s^2 + b s + m g r}, \quad \omega_{n_{ol}} = \sqrt{\frac{m g r}{J_r}}, \quad \zeta_{ol} = \frac{b}{2\sqrt{J_r m g r}}$$

## Proportional feedback (stable equilibrium)

$$H_0(s) = \frac{K G_0(s)}{1 + K G_0(s)} = \frac{K}{J_r s^2 + b s + K + m g r},$$
$$\omega_{n_0} = \sqrt{\frac{K + m g r}{J_r}}, \quad \zeta_0 = \frac{b}{2\sqrt{J_r (K + m g r)}}$$

## Closed-loop stability

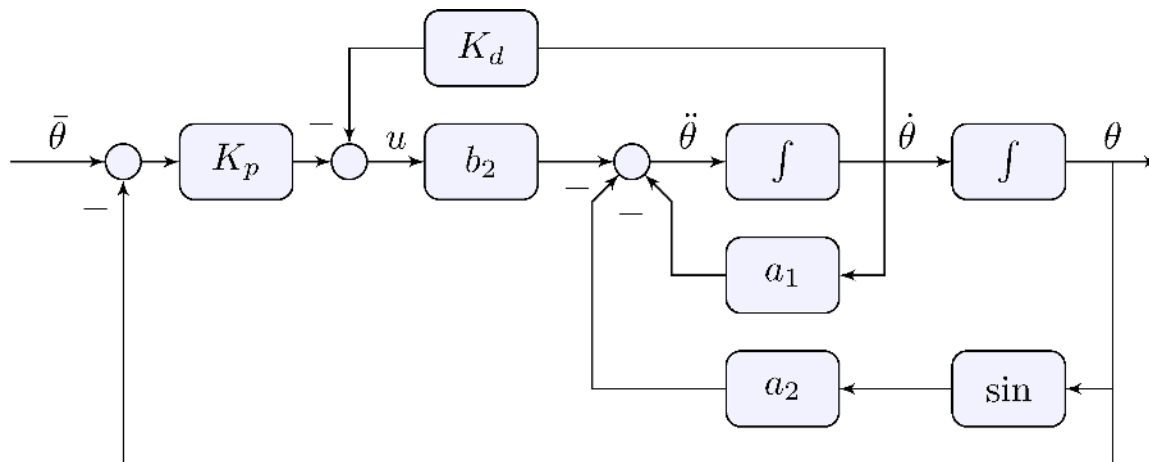
$K > -m g r$ , for example

$$K = 3m g r \quad \implies \quad \omega_{n_0} = 2 \omega_{n_{ol}}, \quad \zeta_0 = \frac{1}{2} \zeta_{ol}$$

## 6.2 Derivative Action

Example: simple pendulum

$$u(t) = K_p \left( \bar{\theta}(t) - \theta(t) \right) - K_d \dot{\theta}(t)$$

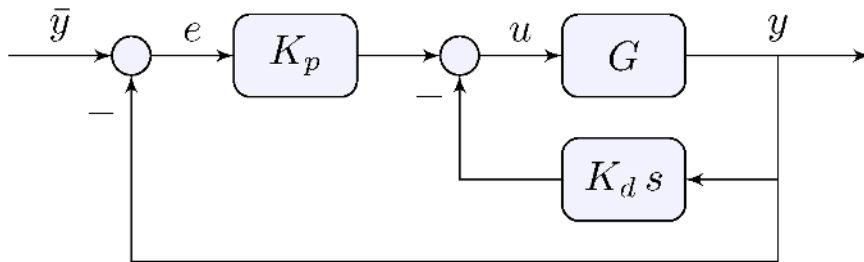


# Proportional-derivative feedback

$$u(t) = K_p (\bar{y}(t) - y(t)) - K_d \dot{y}(t)$$

## Frequency-domain

$$U(s) = K_p E(s) - K_d s Y(s), \quad E(s) = \bar{Y}(s) - Y(s)$$

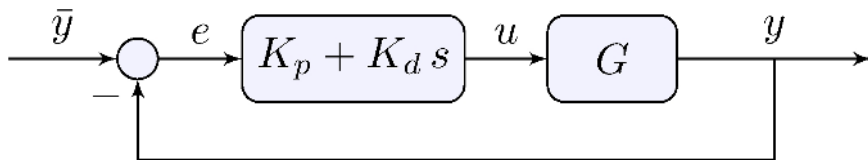


# Proportional-derivative feedback

$$u(t) = K_p e(t) + K_d \dot{e}(t), \quad e(t) = \bar{y}(t) - y(t)$$

## Frequency-domain

$$U(s) = (K_p + sK_d)E(s), \quad E(s) = \bar{Y}(s) - Y(s)$$



- Same if  $\bar{y}$  is constant!
- If not, extra zero

# Example: simple pendulum

## Unstable equilibrium

$$H_{\pi}(s) = \frac{(K_p + K_d s)G_{\pi}(s)}{1 + (K_p + sK_d)G_{\pi}(s)} = \frac{K_p + K_d s}{J_r s^2 + (K_d + b)s + (K_p - m g r)},$$

$$\omega_{n_{\pi}} = \sqrt{\frac{K_p - m g r}{J_r}}, \quad \zeta_{\pi} = \frac{K_d + b}{2\sqrt{J_r(K_p - m g r)}}$$

- Choose  $K_p$  to set desired natural frequency, e.g.

$$K = 3m g r \quad \implies \quad \omega_{n_{\pi}} = \sqrt{2} \omega_{n_{ol}}$$

- Choose  $K_d$  to set desired damping ratio, e.g.  $\zeta_{\pi} = \frac{\sqrt{2}}{2} \approx 0.7$

$$K_d = 2 \zeta_{\pi} \sqrt{J_r(K_p - m g r)} - b = 2\sqrt{J_r m g r} - b$$

- Similarly around stable equilibrium

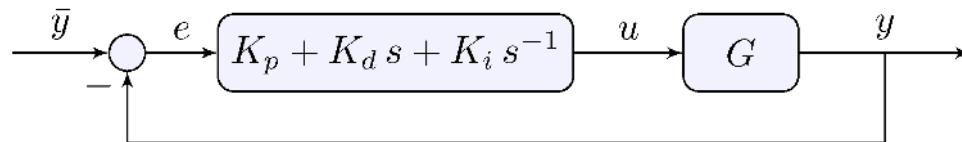
## 6.3 Proportional-Integral-Derivative Control

### Proportional-integral-derivative feedback

$$u(t) = K_p e(t) + K_d \dot{e}(t) + K_i \int_0^t e(\tau) d\tau, \quad e(t) = \bar{y}(t) - y(t)$$

### Frequency-domain

$$U(s) = (K_p + K_d s + K_i s^{-1}) E(s) = \frac{K_d s^2 + K_p s + K_i}{s} E(s), \quad E(s) = \bar{Y}(s) - Y(s)$$



- integral action: asymptotic tracking of constant reference
- derivative action: better “damping”
- Extensive literature on properties of PID control

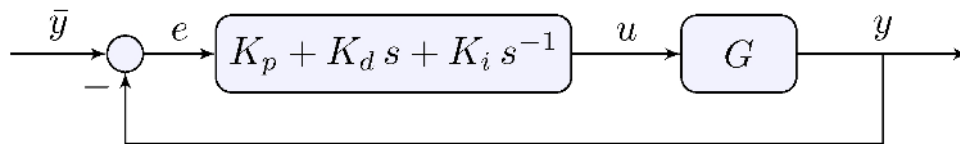


# Proportional-integral-derivative feedback

$$u(t) = K_p e(t) + K_d \dot{e}(t) + K_i \int_0^t e(\tau) d\tau, \quad e(t) = \bar{y}(t) - y(t)$$

## Frequency-domain

$$U(s) = (K_p + K_d s + K_i s^{-1}) E(s) = \frac{K_d s^2 + K_p s + K_i}{s} E(s), \quad E(s) = \bar{Y}(s) - Y(s)$$

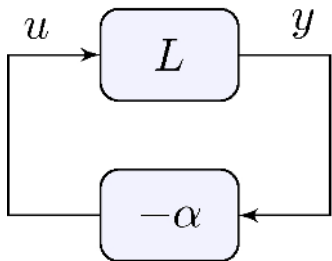


- Transfer-function is not proper: not realizable!
- If  $G$  is second-order then closed-loop transfer-functions are third-order
- Calculation of the characteristic roots is *already* too complicated

## 6.4 Root-Locus

### The basic idea

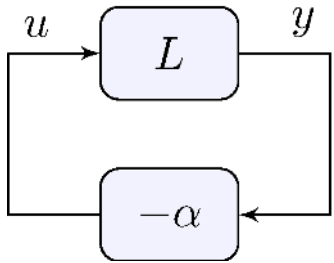
- Can we quickly *sketch* the location of the closed-loop poles without calculating them?
- Root *locus* = location of the roots
- Set of simple rules for sketching the root-locus of the feedback system



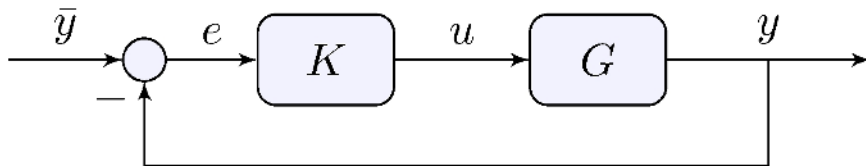
as a function of  $\alpha > 0$

- Make your feedback system look like it and apply the rules
- $L$  is the *loop transfer-function*

## Feedback diagram for root-locus



## Example



## Closed-loop poles

$$1 + \alpha L(s) = 0, \quad K(s) = \alpha C(s), \quad L(s) = C(s)G(s)$$

# Root-locus rules

## Key observation

$$L(s) = -\frac{1}{\alpha} < 0 \quad \implies \quad \angle L(s) = \pi$$

## Rational loop transfer-function

$$L = \beta \frac{N_L}{D_L} = \frac{\beta (s - z_1)(s - z_2) \cdots (s - z_m)}{(s - p_1)(s - p_2) \cdots (s - p_n)}$$

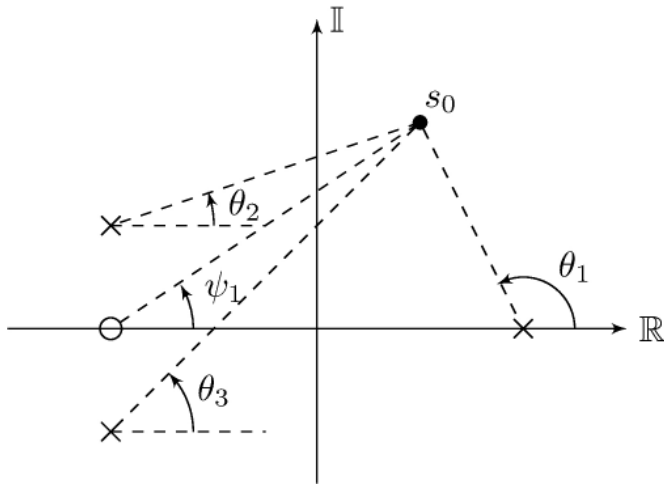
then

$$\angle L = \angle \beta + \angle N_L - \angle D_L = \angle \beta + \sum_{i=1}^m \angle(s - z_i) - \sum_{k=1}^n \angle(s - p_k)$$

# Root-locus rules

Phase can be calculated graphically

$$\begin{aligned}\angle L(s_0) &= \angle \beta + \sum_{i=1}^m \angle(s_0 - z_i) - \sum_{k=1}^n \angle(s_0 - p_k) \\ &= \angle \beta + \psi_1 - \theta_1 - \theta_2 - \theta_3\end{aligned}$$



# Root-locus rules

## Assumption

$L$  rational and strictly proper with real coefficients

## Rules

- Root-locus is a set of  $n$  continuous curves
- The curves:
  - begin at the poles of  $L$

$$\lim_{\alpha \rightarrow 0} D_L(s) (1 + \alpha L(s)) = \lim_{\alpha \rightarrow 0} D_L(s) + \alpha \beta N_L(s) = D_L(s)$$

- end at the zeros of  $L$

$$\lim_{\alpha \rightarrow \infty} \alpha^{-1} D_L(s) (1 + \alpha L(s)) = \lim_{\alpha \rightarrow \infty} \alpha^{-1} D_L(s) + \beta N_L(s) = \beta N_L(s)$$

- $n - m \geq 1$  curves go to infinity

# Root-locus rules

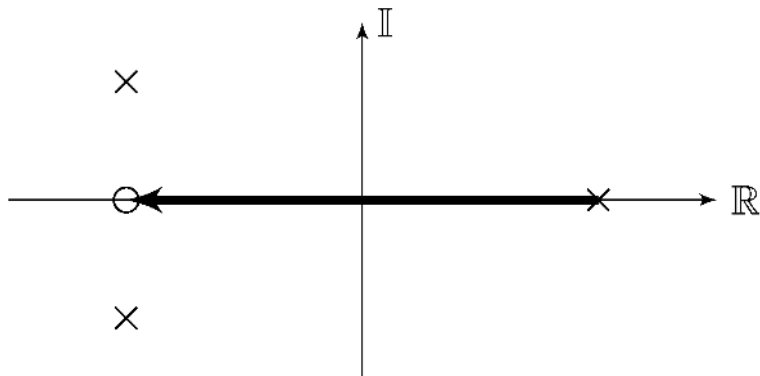
## Assumption

$L$  rational and strictly proper with real coefficients and  $\beta > 0$

## Rule

- $s \in \mathbb{R}$  is in the root-locus iff  $L$  has an odd number of poles and zeros with real part greater than or equal to  $s$

## Example (complex locus not shown)



- What would the  $\beta < 0$  version be?

# Root-locus rules

## Assumption

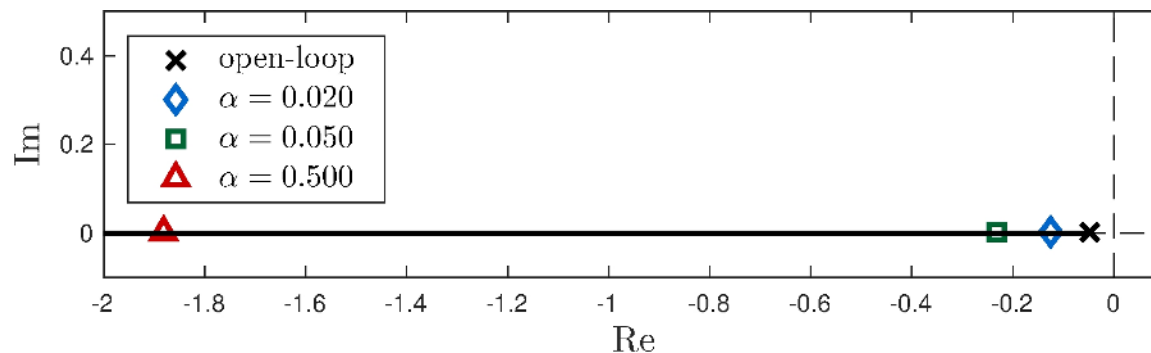
$L$  rational and strictly proper with real coefficients and  $\beta > 0$

## Rule

- $s \in \mathbb{R}$  is in the root-locus iff  $L$  has an odd number of poles and zeros with real part greater than or equal to  $s$

## Example: car model under proportional feedback

$$\alpha = K_p, \quad L(s) = \frac{\frac{p}{m}}{s + \frac{b}{m}} = \frac{3.7}{s + 0.05}$$





# Root-locus rules

## Assumption

$L$  rational and strictly proper with real coefficients

## Rule

- If  $n > m + 1$ , then  $n - m$  roots converge to  $n - m$  straight-line asymptotes intersecting at:

$$c = \frac{\sum_{k=1}^n p_k - \sum_{i=1}^m z_i}{n - m} \in \mathbb{R}$$

- If  $\beta > 0$ , the asymptotes intersect the real axis at angles

$$\phi_k = \frac{\pi + k 2\pi}{n - m}, \quad k = 0, \dots, n - m - 1$$

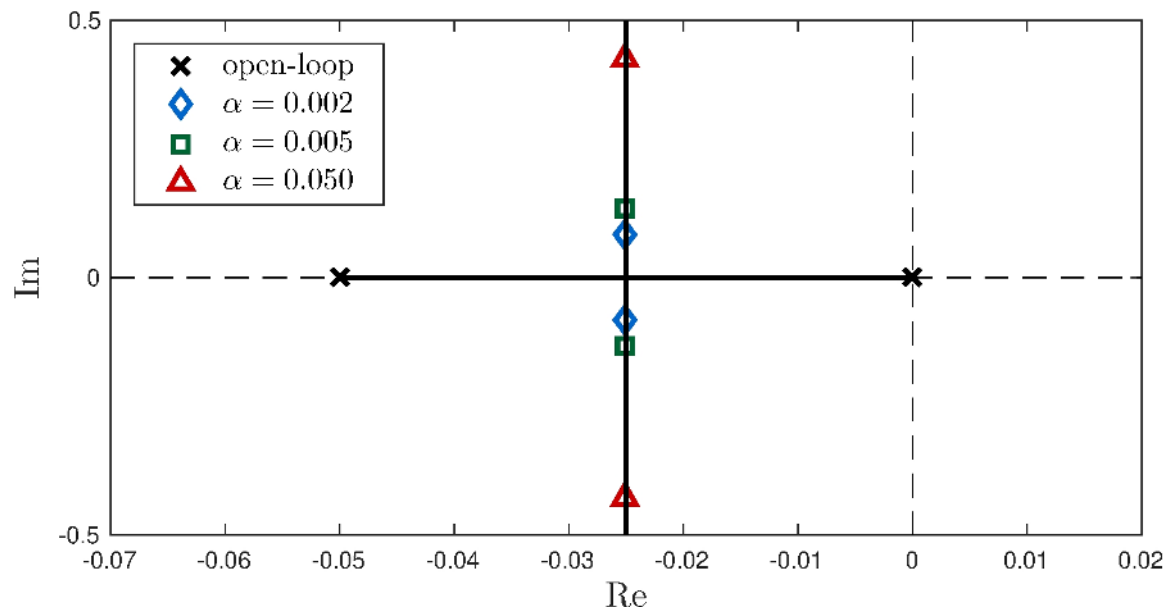
# Root-locus rules

## Example: car model under integral feedback

$$\alpha = K_i, \quad L(s) = G(s) C(s) = \frac{3.7}{s(s + 0.05)}$$

### Asymptotes

$$c = \frac{-0.05 + 0}{2} = -0.025, \quad \phi_1 = \frac{\pi}{2}, \quad \phi_2 = \frac{3\pi}{2}$$



# Root-locus rules

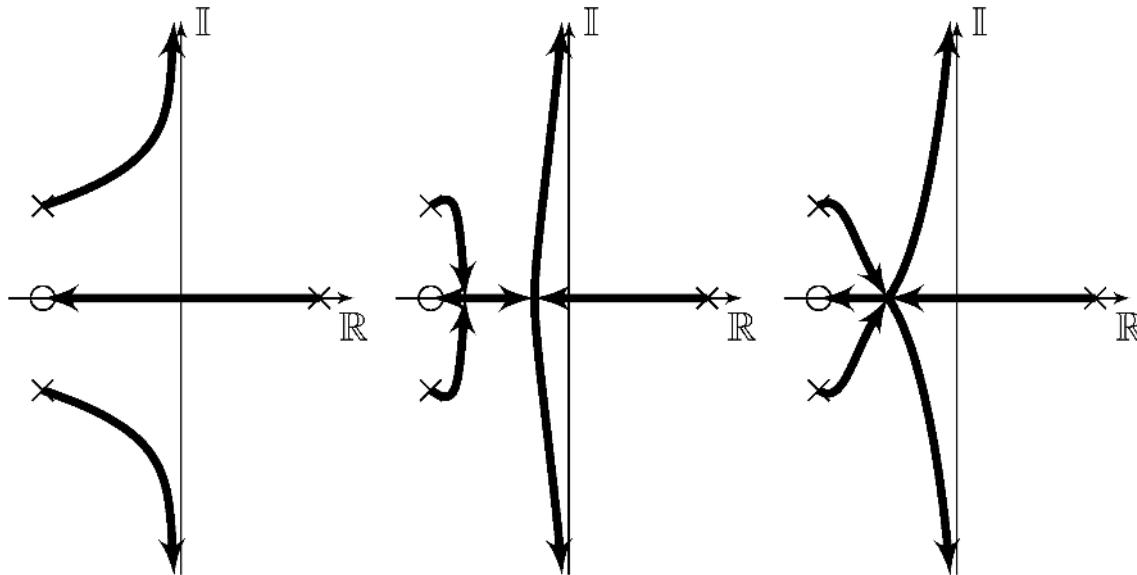
## Example: model with three poles and one zero

$$p_1 = a, \quad p_2 = -a + jb, \quad p_3 = -a - jb, \quad z_1 = -a$$

## 2 asymptotes

$$c = \frac{(-a - jb) + (-a + jb) + (a) - (-a)}{3 - 1} = 0, \quad \phi_1 = \frac{\pi}{2}, \quad \phi_2 = \frac{3\pi}{2}$$

## Possible root-loci



## Root-locus rules

Example: car model under proportional-integral feedback

$$G = \frac{\frac{p}{m}}{s + \frac{b}{m}} = \frac{3.7}{s + 0.05}, \quad K(s) = K_p + \frac{K_i}{s} = K_p \frac{s + K_i/K_p}{s}$$

Loop transfer-function

$$\alpha = K_p, \quad L = G C = \frac{3.7 (s + K_i/K_p)}{s (s + 0.05)}$$

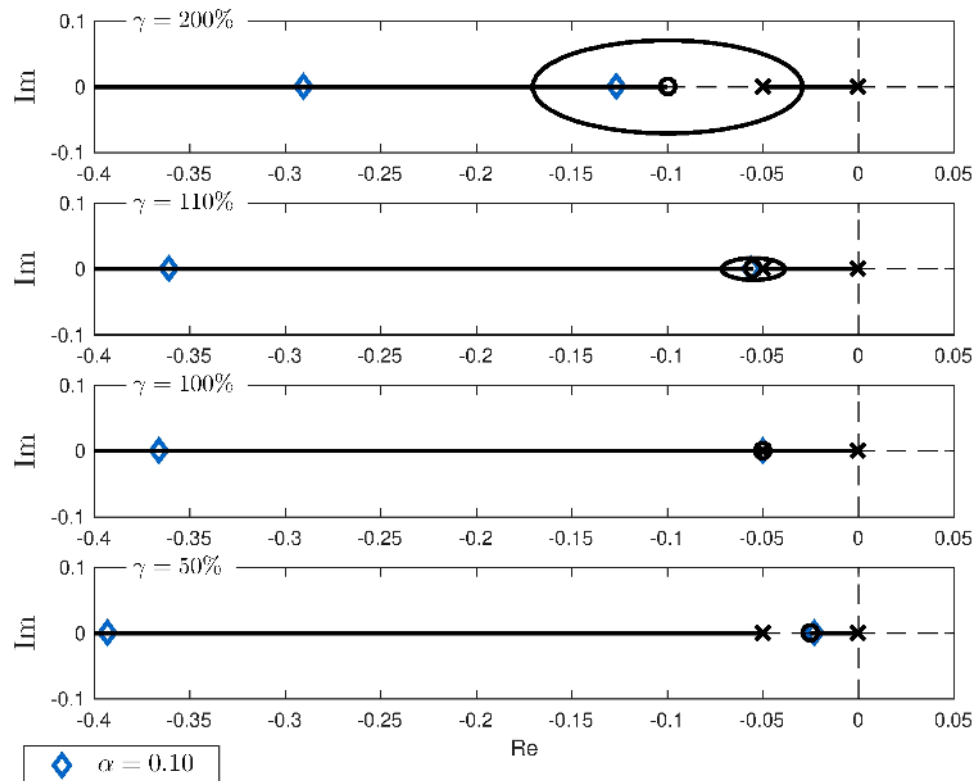
Choice of controller zero

$$\frac{K_i}{K_p} = \gamma \frac{b}{m} = \gamma 0.05$$

- $\gamma = 1$  is a pole-zero cancellation

# Root-locus rules

## Example: car model under proportional-integral feedback



# 6.5 Control of the Simple Pendulum - Part I

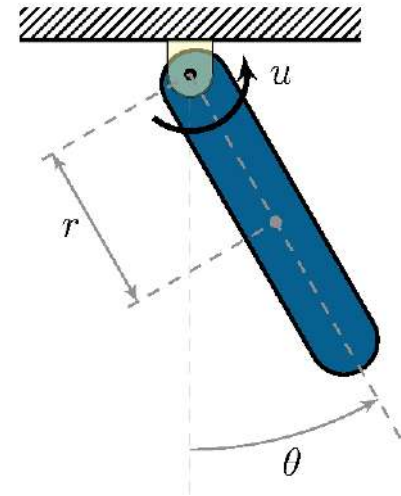
## Proportional-derivative feedback

### Controller

$$C_{PD}(s) = K_d (s + z), \quad z = \frac{K_p}{K_d}$$

### Data

$$\begin{aligned} m &= 0.5\text{kg}, & \ell &= 0.3\text{m}, & r &= \ell/2 = 0.15\text{m} \\ b &= 0\text{kg/s}, & g &= 9.8\text{m/s}^2, & J &= \frac{m\ell^2}{12} = 3.75 \times 10^{-3}\text{kg m}^2, & J_r &= J + mr^2 = 0.015 \end{aligned}$$



### Gains

$$K_d = 3 m g r \approx 0.21, \quad K_p = 2\sqrt{J_r m g r} - b \approx 2.21, \quad z \approx 10.5, \quad C_{PD}(s) \approx 0.21(s + 10.5)$$

# Proportional-derivative feedback

## System model

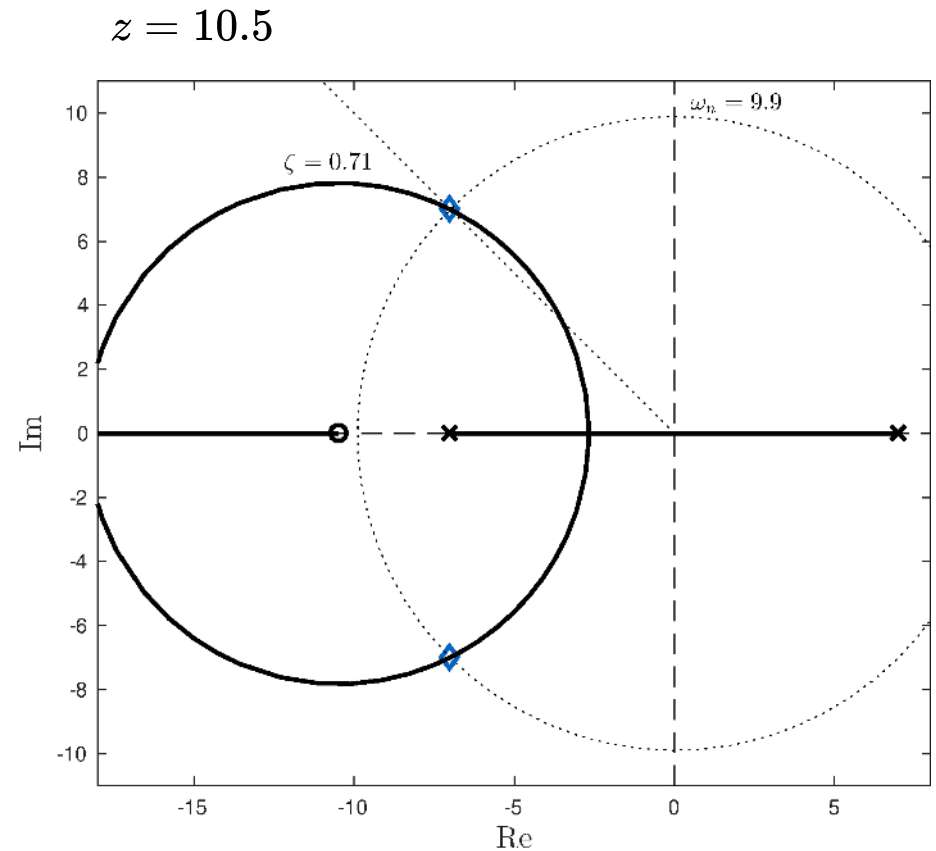
$$G_{\pi}(s) = \frac{66.7}{(s + 7)(s - 7)}$$

## Controller

$$K(s) = C_{PD}(s) = K_d(s + z), \quad z = \frac{K_p}{K_d} > 0$$

## Root-locus

$$\alpha = K_d, \quad L = \frac{1}{K_d} G_{\pi} C_{PD} = \frac{66.7(s + z)}{(s + 7)(s - 7)}$$



- Diamonds:  $K_d \approx 0.21 \implies \omega_{n_{\pi}} = \sqrt{2} \omega_{n_{ol}} \approx 9.90, \quad \zeta_{\pi} = 1/\sqrt{2} \approx 0.71$
- PD controller is not proper, hence not realizable: we need a pole
- How about a pole at the origin for some desirable integral action?

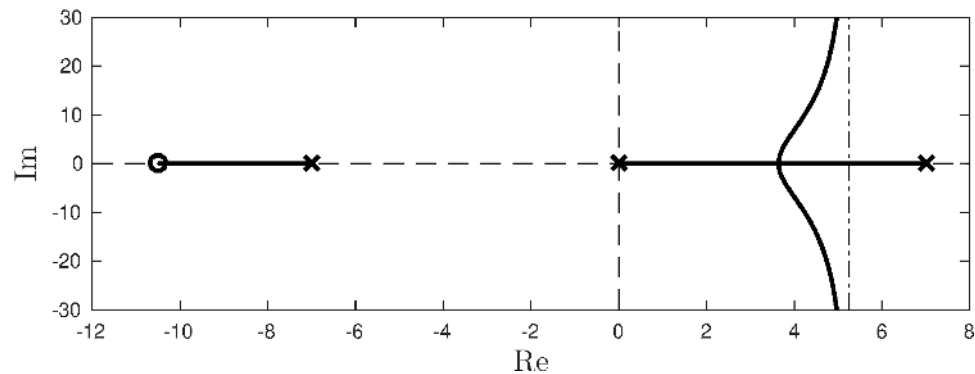
# Proportional-integral feedback

## Controller

$$K = K_p \frac{(s + z)}{s}$$

## Root-locus

$$\alpha = K_p, \quad L = \frac{66.7(s + z)}{s(s + 7)(s - 7)}$$



- No  $K_p > 0$  can make closed-loop stable!



# Proportional-derivative feedback plus pole Controller

$$K(s) = K \frac{s + z}{s + p}$$

## Root-locus

$$\alpha = K \quad L = \frac{66.7(s + z)}{(s + 7)(s - 7)(s + p)}$$

## 2 asymptotes

$$c = \frac{7 - 7 + (-p) - (-z)}{n - m} = \frac{z - p}{2}, \quad \phi_1 = \frac{\pi}{2}, \quad \phi_2 = \frac{3\pi}{2}$$

## Stability

Possible with large  $K$  only if

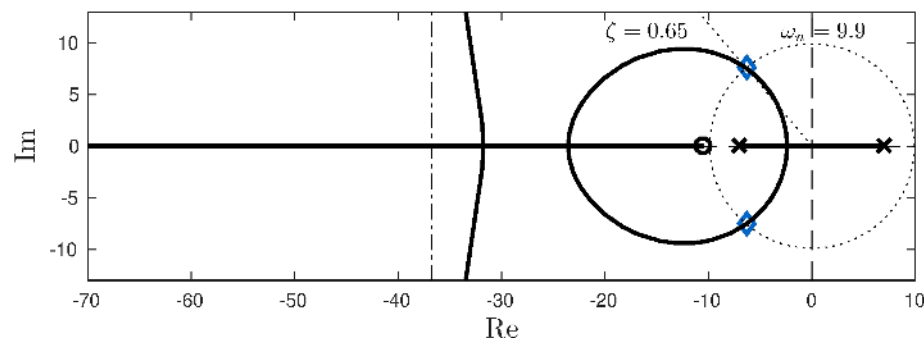
$$c < 0 \quad \implies \quad p > z$$

# Proportional-derivative feedback plus pole

## Root-locus

$$\alpha = K \quad L = \frac{66.7(s + z)}{(s + 7)(s - 7)(s + p)}$$

## Choice of pole



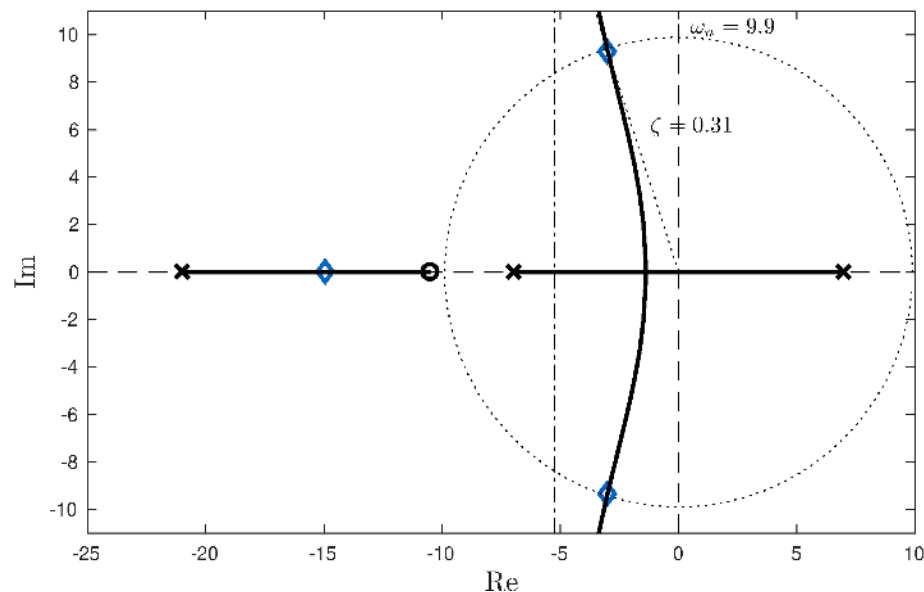
- $p = 8z \gg z \approx 10.5$
- Root-locus near origin is mostly unaffected

# Proportional-derivative feedback plus pole

## Root-locus

$$\alpha = K \quad L = \frac{66.7(s + z)}{(s + 7)(s - 7)(s + p)}$$

## Choice of pole



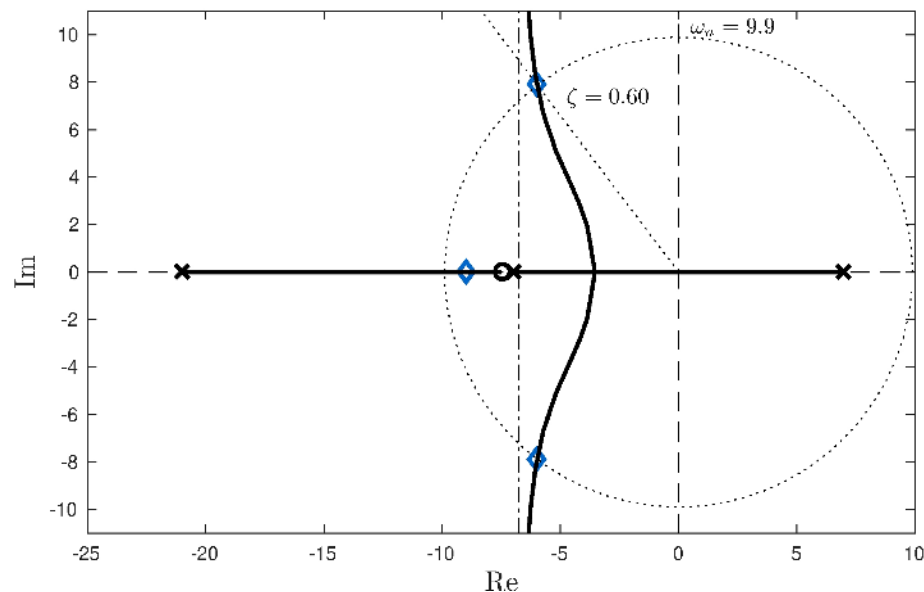
- $p = 2z > z \approx 10.5$
- Asymptote interferes with root-locus near the origin
- Low damping

# Proportional-derivative feedback plus pole

## Root-locus

$$\alpha = K \quad L = \frac{66.7(s + z)}{(s + 7)(s - 7)(s + p)}$$

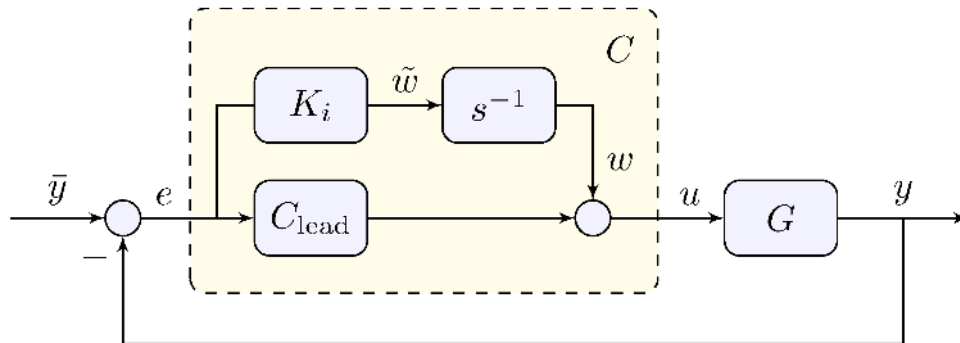
## Choice of pole and adjustment of zero



- $p = 2z > \tilde{z} \approx 7.5$
- Adjusting zero improves damping
- Final controller

$$C_{\text{lead}}(s) \approx 3.83 \frac{s + 7.5}{s + 21}$$

## Can we add integral action?



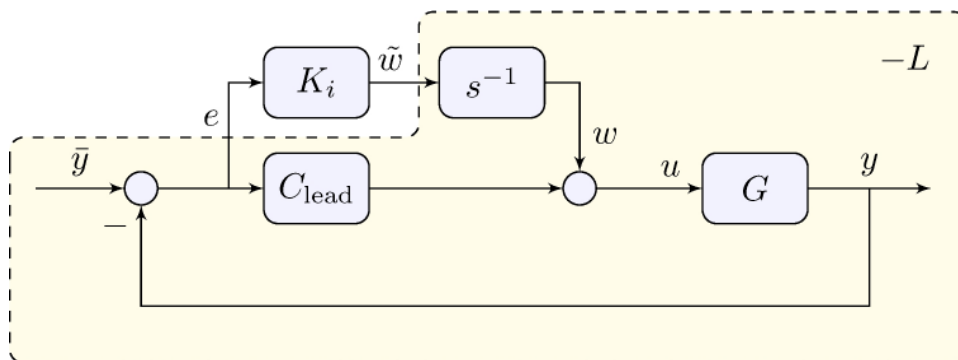
- Previous attempt failed to stabilize
- Additive approach means that stability is guaranteed for  $K_i$  small
- Structure of controller

$$C(s) = \frac{K_i}{s} + C_{\text{lead}}(s) = K \frac{s^2 + (z + K_i/K)s + pK_i/K}{s(s + p)}$$

- What is  $L$ ?

# Can we add integral action?

## Root-locus analysis



## Loop transfer-function

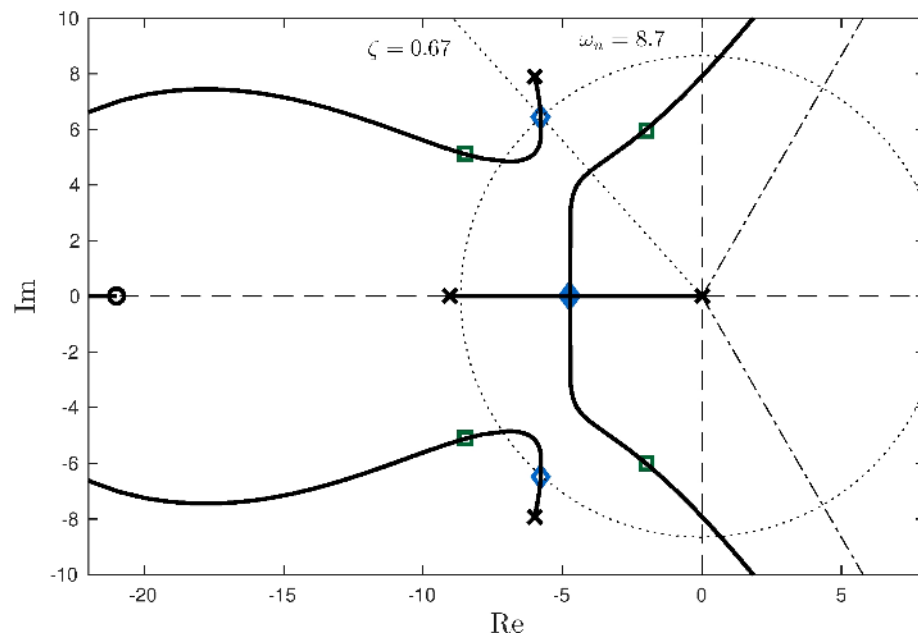
$$\alpha = K_i, \quad L = -\frac{E}{\tilde{W}} = -s^{-1} \frac{E}{W} = s^{-1} D = \frac{s^{-1} G_\pi}{1 + G_\pi C_{\text{lead}}}$$

# Can we add integral action?

## Root-locus analysis

$$\alpha = K_i, \quad L = -\frac{E}{\tilde{W}} = -s^{-1} \frac{E}{W} = s^{-1} D = \frac{s^{-1} G_\pi}{1 + G_\pi C_{\text{lead}}}$$

## Choice of integral gain



- Diamonds: double real poles,  $\omega_n \approx 8.7$
- Squares:  $\omega_n \approx 9.9$ , underdamped poles
- Final controller:

$$C_6(s) \approx 3.83 \frac{(s + 6.86)(s + 0.96)}{s(s + 21)}$$

- See book for:
  - controller design around the stable equilibrium
  - direct root-locus design with integrator

# Learn more

- [Book website](#)